

LLM PROJECT

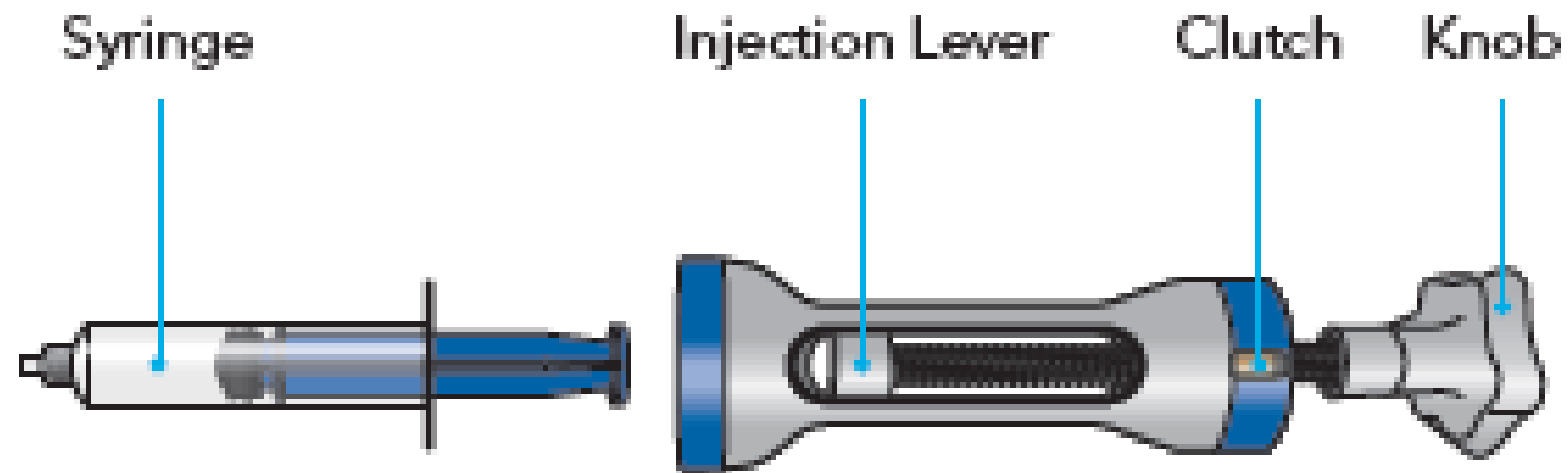
SYNTHETIC DATA GENERATION PIPELINE FOR MEDICAL ABSA

Lecturer: Dr. Aparcin Alexander

Students: Shoham Rachel Amira & Adi Malka

Background

This project focuses on building an LLM-based model that analyzes patient reviews about the Uroject12 syringe device and identifies whether different aspects of the experience are described positively or negatively.





Problem Definition



motivation

A major challenge in medical ABSA is the lack of large labeled datasets, which motivated the creation of a controlled synthetic review generation pipeline.

Why is it important?

Patient feedback can provide valuable insights about usability, comfort, safety, and overall satisfaction with medical devices.

Main challenge

Patient reviews are written in free text, which makes it difficult to manually extract consistent and structured insights from large amounts of feedback.

Current Solutions

Today, feedback analysis is often performed manually or with simple keyword-based methods that may miss context and sentiment.

Previous work

Paper	Task	Methods / Models	Main Results
Revealing Patient Dissatisfaction With Health Care Resource Allocation in Multiple Dimensions Using Large Language Models and the International Classification of Diseases 11th Revision: Aspect-Based Sentiment Analysis	Aspect-Based Sentiment Analysis (ABSA) on patient reviews	ChatGPT with aspect-based prompting	Demonstrated that ChatGPT can identify dissatisfaction related to specific healthcare aspects rather than only overall sentiment.
Using Large Language Models for sentiment analysis of health-related social media data: empirical evaluation and practical tips	Sentiment analysis of health-related social media text	GPT-3.5-Turbo, FLAN-T5, and BERT-based models	Showed that LLM performance on healthcare sentiment analysis strongly depends on prompting, fine-tuning, and validation strategies.
Large language models and synthetic health data: progress and prospects	Review of synthetic healthcare data generation	Review of LLM-based synthetic data generation methods	Highlighted the potential of LLMs for generating synthetic health data while emphasizing the importance of validation, privacy, and quality control.



Project task description

- **Problem Definition**

The project aims to build an LLM-based model that performs aspect-based sentiment analysis on patient reviews about the Uroject12 syringe device.

- **Input**

The input is a free-text patient review describing the user's experience with the device.

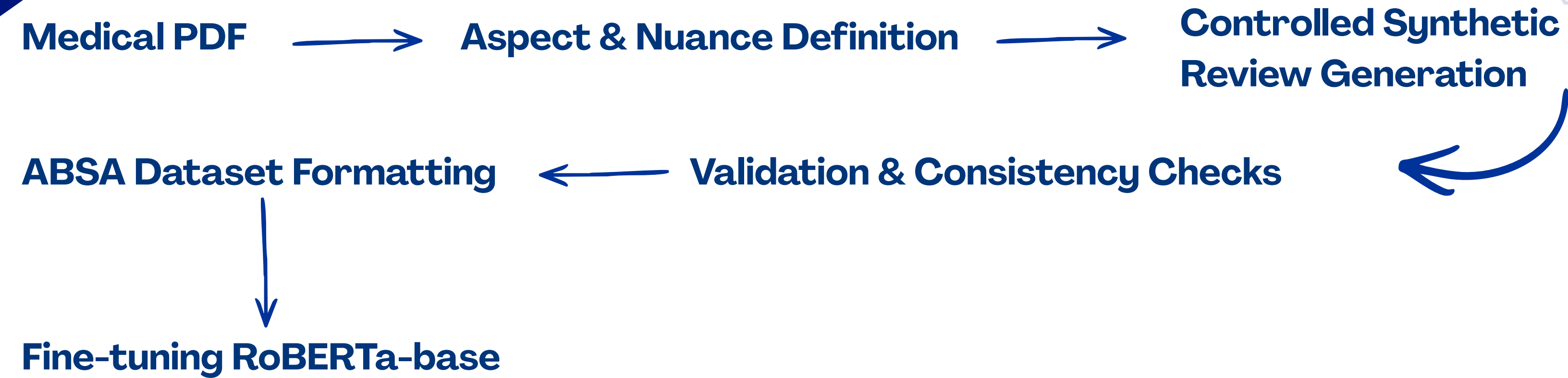
- **Output**

The output is a sentiment classification (positive, negative, neutral, or mixed) for each relevant aspect mentioned in the review, such as ease of use, cleaning and sterilization, lever operation, syringe compatibility, and device durability.

- **Project Novelty**

Unlike traditional keyword-based sentiment analysis, the proposed LLM approach can better understand context, identify multiple aspects within the same review, and capture more nuanced opinions.

Processing Pipeline & Dataset



Dataset

- 10,000 synthetic reviews
- 24,921 aspect-level samples
- 8 predefined aspects
- 4 sentiment classes

Validation

- JSON validation
- Duplicate detection
- Label consistency checks
- Sentiment Class Distribution Analysis
 - negative: 6,275
 - positive: 6,261
 - neutral: 6,230
 - mixed: 6,155
- Aspect Distribution Analysis
 - ~3,080–3,140 samples per aspect
- Manual human review sample
- LLM consistency audit
 - 95.16% consistency rate

Model & Fine-Tuning

Baseline Model

- Pretrained RoBERTa-base Transformer
- Fine-tuned for Aspect-Based Sentiment Analysis
- Predicts sentiment for a specific aspect within a review

Example Input

- Aspect: inspection_and_maintenance
“inspecting the device before each use is important, especially to check for signs of wear or any operational issues, but this step is routine and manageable.”
- Predicted sentiment:
Neutral

Fine-Tuning Strategy

- Stratified train/validation/test split
- Dynamic padding
- Early stopping
- Macro-F1 model selection
- Initial training on 2,000 samples
- batch-size: 8
- learning rate: $1e-5$
- epochs: 6

Evaluation & Results

Classification Report

	precision	recall	f1-score	support
negative	0.89	0.99	0.93	79
neutral	0.90	0.93	0.92	71
positive	0.95	0.97	0.96	74
mixed	0.97	0.80	0.88	76
accuracy			0.92	300
macro avg	0.93	0.92	0.92	300
weighted avg	0.93	0.92	0.92	300

Confusion Matrix

```
[[ 78  0  1  0 ]
 [ 1 66  2  2 ]
 [ 0  2 72  0 ]
 [ 9  5  1 61 ]]
```

Main Errors

- Mixed ↔ Neutral
- Mixed ↔ Positive/Negative

Reason

Mixed sentiment contains both positive and negative expressions.

Future Work

- Compare with an LLM-based approach: Phi-3
- Train on larger datasets: 2,000 ABSA rows → 24,921 ABSA rows
- Improve mixed sentiment detection: tune class weights, learning rate, number of epochs, and decision thresholds
- Add automatic aspect detection for raw free-text reviews

References



1. Li J, Yang Y, Mao C, Pang PC, Zhu Q, Xu D, Wang Y. Revealing Patient Dissatisfaction With Health Care Resource Allocation in Multiple Dimensions Using Large Language Models and the International Classification of Diseases 11th Revision: Aspect-Based Sentiment Analysis. J Med Internet Res. 2025 Mar 17;27:e66344. doi: 10.2196/66344. PMID: 40096682; PMCID: PMC11959199.
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2. He L, Omranian S, McRoy S, Zheng K. Using Large Language Models for sentiment analysis of health-related social media data: empirical evaluation and practical tips. AMIA Annu Symp Proc. 2025 May 22;2024:503–512. PMID: 40417506; PMCID: PMC12099340.
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3. Smolyak D, Bjarnadóttir MV, Crowley K, Agarwal R. Large language models and synthetic health data: progress and prospects. JAMIA Open. 2024 Oct 26;7(4):ooae114. doi: 10.1093/jamiaopen/ooae114. PMID: 39464796; PMCID: PMC11512648.
<https://pubmed.ncbi.nlm.nih.gov/39464796/>

The slide features decorative geometric patterns in the corners. The top-left corner has a dark blue triangle pointing right, partially overlapping a light blue triangle. The top-right corner has a complex pattern of overlapping hexagons and lines in light blue and dark blue. The bottom-left corner has a similar hexagonal pattern in light blue and dark blue. The bottom-right corner has a dark blue triangle pointing left, partially overlapping a light blue triangle.

Thank You